SIS 3204Y(5) – Natural Language Processing

Week 1 – Revision Questions and Answers

Based on Lecture 1 notes, aligned with allocated question marks. Concise technical descriptions using simple language.

Question 1: Define Natural Language Processing (NLP). — 2 Marks

- **Natural Language Processing (NLP)** is a branch of Artificial Intelligence (AI) that enables computers to understand, interpret, generate, and interact using human language in text or speech.

Question 2: State FOUR real-world applications of NLP. — 4 Marks

- **Chatbots and virtual assistants:** Answer questions and communicate naturally with users.
- **Machine translation:** Translates text or speech seamlessly from one language to another.
- **Email spam filtering:** Identifies, categorises, and filters unwanted or malicious emails.
- **Sentiment analysis:** Detects opinions and emotions in text, such as classifying reviews as positive or negative.

Question 3: Explain why human language is difficult for computers to understand. — 6 Marks

- **Ambiguity:** The same word or sentence can have multiple meanings. For example, *bank* can refer to a financial institution or the side of a river.
- **Context dependence:** Meaning relies heavily on the specific situation and surrounding words.
- **Sarcasm and irony:** Spoken or written words may express the exact opposite of their literal meaning.
- **Idioms and figurative language:** Expressions like *"break a leg"* cannot be understood literally.
- **Language variations:** People frequently use slang, abbreviations, typos, and distinct regional dialects.
- **Emotions and cultural differences:** Tone, underlying feelings, and background knowledge heavily influence interpretation.

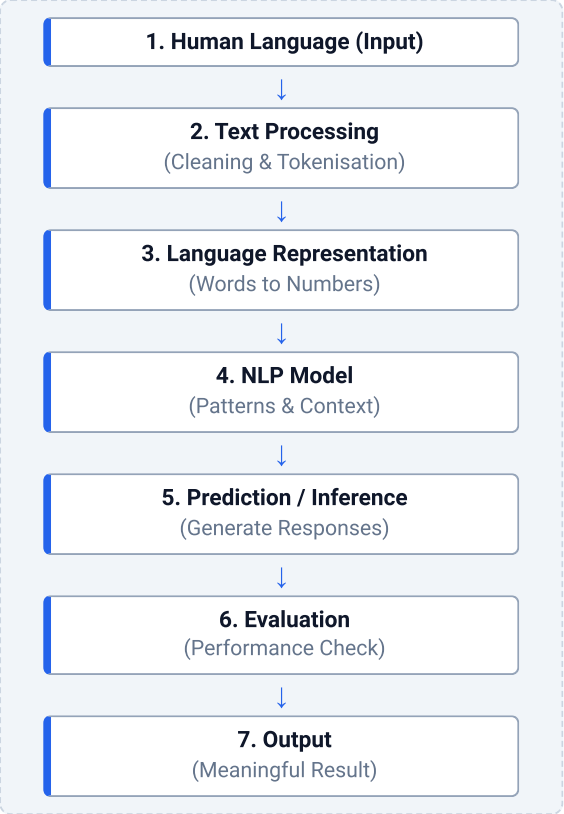
Question 4: Differentiate between Traditional NLP and Modern NLP. — 6 Marks

Traditional NLP	Modern NLP
Uses handcrafted rules created by experts.	Uses data-driven models that learn from examples.
Works with small, task-specific datasets.	Learns from massive amounts of text data.
Requires separate dedicated systems for different tasks.	General-purpose models perform many diverse tasks.
Has limited context understanding.	Understands long-range context & complex relationships.
Struggles with ambiguity and language variations.	Handles ambiguity and language variations effectively.
Difficult to scale due to growing rule complexity.	Highly scalable; improves with more data & compute.

Examples: Traditional NLP includes rule-based spell/grammar checkers; Modern NLP includes ChatGPT, Claude, and Gemini.

Question 5: Describe the NLP pipeline using a labelled diagram. — 6 Marks

The NLP pipeline is a sequence of structured steps that converts raw human language into meaningful machine outputs.



Pipeline Step Explanations:

- **Human Language (Input):** Raw text or speech collected from users or datasets.
- **Text Processing:** Text is cleaned, normalised, and split into smaller units (tokens).
- **Language Representation:** Converts words/sentences into vectors (e.g., word embeddings).
- **NLP Model:** Learns linguistic patterns, relationships, and context from data.
- **Prediction / Inference:** Applies learned knowledge to output predictions or responses.
- **Evaluation:** Checks model accuracy using task-specific metrics.
- **Output:** Final result delivered to the user (e.g., summary, answer, translation).

Question 6: What is tokenisation? Give one suitable example. — 3 Marks

- **Definition:** Tokenisation is the process of splitting text into smaller units called *tokens*, such as words, subwords, or punctuation marks.
- **Importance:** It is a critical initial text processing step that prepares raw text for analysis by downstream computational models.
- **Example:**

  Input: "I love NLP"
  Output: `["I", "love", "NLP"]`

Question 7: Explain the role of word embeddings in Natural Language Processing. — 4 Marks

- **Definition:** Word embeddings represent words as real-valued numerical vectors in a continuous vector space that computers can compute upon.
- **Semantic Capture:** They help NLP models capture semantic meaning, grammatical traits, and relationships between words.
- **Vector Proximity:** Words with similar contextual meanings are mapped to vector positions that are geographically closer together in vector space.
- **Impact:** Enables modern models to process language based on deeper contextual relationships rather than superficial keyword matching.
- **Example:** Words like *"king"* and *"queen"* share similar vector representations due to related semantic contexts

Question 8: Describe the evolution of NLP from rule-based systems to LLMs. 7 Marks

- **Rule-based systems (1950s–1980s):** Relied on expert-crafted grammar rules, dictionaries, and pattern matching. Simple, but struggled with ambiguity.
- **Statistical NLP (1990s):** Applied statistical probability to large text corpora (e.g., N-gram language models).
- **Machine Learning (early 2000s):** Models learned patterns directly from labelled training data (e.g., Naive Bayes for spam detection).
- **Deep Learning (2000s–early 2010s):** Neural networks (RNNs, LSTMs) automated feature extraction and sequence modeling.
- **Word Embeddings:** Vector techniques (Word2Vec) mapped semantic word meanings mathematically.
- **Transformers (2017 onwards):** Introduced self-attention mechanism, processing sequence elements in parallel with full context awareness.
- **Large Language Models (2020 onwards):** Massive multi-task models trained on huge text datasets (e.g., ChatGPT, Claude, Gemini).

Question 9: What is a Transformer and why is it important in modern NLP? 5 Marks

- **Definition:** A neural network architecture designed specifically to process and model sequential language data.
- **Attention Mechanism:** Utilises *self-attention* to dynamically compute relationships between all words in a sentence simultaneously.
- **Long-Range Dependencies:** Captures global context significantly better than traditional recurrent architectures (like RNNs/LSTMs).
- **Parallel Processing:** Allows computational operations across input tokens to run in parallel, dramatically improving training efficiency.
- **Foundation of Modern AI:** Serves as the core architectural backbone for flagship models such as BERT, GPT, and modern LLMs.

Question 10: What are Large Language Models (LLMs)? Give TWO examples. 4 Marks

- **Definition:** Advanced deep learning models trained on massive volumes of text data to comprehend, reason, and generate human-like language.
- **Capabilities:** They learn deep probabilistic patterns to perform diverse tasks like Q&A, summarisation, coding, and dialogue.
- **Examples: 1.** ChatGPT (OpenAI) | **2.** Claude (Anthropic)

Question 11: Explain Retrieval-Augmented Generation (RAG) and its purpose. 5 Marks

- **Concept:** An architecture combining external *information retrieval* with generative language models.
- **Process:** Fetches factual documents from an external repository and passes them as explicit context to the generator model.
- **Purpose & Benefits:**
  - Ensures answers are grounded in verifiable, domain-specific, or up-to-date data.
  - Significantly mitigates model **hallucinations** and increases output reliability.

Question 12: What is hallucination in a Large Language Model? 3 Marks

- **Definition:** A phenomenon where an LLM generates output that is factually incorrect, ungrounded, or completely fabricated.
- **Characteristics:** The generated text often sounds authoritative, logical, and convincing despite being entirely false.
- **Example:** Inventing non-existent academic references, historical facts, or technical statistics.

Question 13: Discuss THREE ethical challenges associated with NLP systems. 6 Marks

- **1. Bias and Fairness:** Models inherit social biases present in training data, resulting in unfair outputs or demographic stereotypes.
- **2. Privacy and Security:** Systems may accidentally memorise and leak sensitive user information or personal confidential records.
- **3. Misinformation & Hallucination:** Convincing but incorrect outputs can lead to rapid spread of false information if left unchecked.

Question 14: Comprehensive Evolution, Benefits, and Challenges of NLP. 15 Marks

1–6. Structural Evolution of NLP Paradigms

- **Rule-Based (50s–80s):** Used handcrafted grammar rules (e.g., spell checkers). *Limitation:* Poor scaling & context handling.
- **Statistical (90s) & Machine Learning (00s):** Applied probabilities (N-grams) and supervised algorithms (Spam filters). *Benefit:* Learned from data. *Challenge:* Heavy feature engineering.
- **Deep Learning & Embeddings (10s):** RNNs/LSTMs & Word2Vec automated representations. Improved sequence tracking.
- **Transformers & LLMs (2017+):** Attention mechanism enabled massive parallel scaling (ChatGPT, Claude, Gemini). Unified broad tasks into single general-purpose engines.

7. Key Benefits

- **High Accuracy:** Deep understanding of context.
- **Versatility:** One general model covers Q&A, translation, and summary.
- **Natural Generation:** Fluid, human-like dialogue.
- **Broad Reach:** Drives healthcare, finance, & education.

8. Key Challenges

- **Hallucinations:** Fabricated facts sounding convincing.
- **Bias & Privacy:** Risk of discriminatory or leaked data.
- **High Compute Cost:** Requires immense GPU resources.
- **Reasoning Gaps:** Struggles with multi-step logic.

Conclusion: NLP transitioned from rigid manual rules to scalable LLMs. Responsible deployment is imperative to guarantee fairness, factual grounding, and security.